Testing the Fluorescence Advantage: Solar-Induced Fluorescence as an Early Indicator of Agricultural Drought Stress in Marathwada, Maharashtra

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Abstract

Drought detection in India, both for official declaration purposes and for triggering payouts under the Pradhan Mantri Fasal Bima Yojana (PMFBY) crop-insurance scheme, still leans heavily on rainfall-deficit records and the Normalized Difference Vegetation Index (NDVI) — two signals that only move once a crop has already begun to visibly suffer. This study asks whether Solar-Induced Fluorescence (SIF), a satellite-derived proxy for photosynthetic activity, picks up that stress measurably earlier than NDVI does. Using GOSIF v2 fluorescence data alongside cloud-screened MODIS NDVI over the eight districts of Marathwada, Maharashtra, across eight growing seasons (2015–2023, excluding 2021 for data-availability reasons), I calculated the lag between the two indices' post-peak seasonal decline. A twenty-year (2001–2020) rainfall climatology puts only two of those eight years past this study's own drought threshold — 2015 (−21.5%) and 2018 (−18.3%) — with the remaining six sitting within roughly one standard deviation of normal. Under the threshold-crossing method, SIF's decline led NDVI's in seven of the eight years; the eighth, 2018, came out essentially flat and marginally negative (−1.1 days) — a real exception, not a rounding artifact. H3, which predicted that drought conditions would widen this lag, still doesn't hold at the larger sample: the two drought years averaged a shorter lag (7.6 days) than the six normal years (15.0 days). A second, independently computed cross-correlation lag backs the direction of this finding without matching its exact numbers — the correlation-maximizing lag was never negative across all eight years, and strictly positive in four of them, with the other four (2018 included) landing at zero. A case-resampling bootstrap (2,000 replicates per year) sharpens that same, more modest claim: the estimated lag sat at or above zero in 100% of replicates in every year, even though its precise day count remains uncertain and no two years' estimates are statistically distinguishable from each other. At the district level, mean SIF and rainfall anomaly correlate significantly across all 64 district-year observations (8 districts × 8 years): Pearson $r = 0.567$, Spearman $\rho = 0.551$, both $p < 0.0001$ — real and still significant, but considerably weaker than the $r = 0.837$ the original 24-point, 3-year sample produced, which looks in hindsight like an artifact of a small, drought-heavy sample. Moran's I confirms genuine spatial structure in both variables, though less consistently for SIF (significant in 4 of 8 years) than for rainfall (significant in all 8), and the correlation's effective independent sample size sits closer to the number of years than the number of district-year rows. Checking against the one well-documented official drought declaration available (Maharashtra, 31 October 2018) turned up a reversal from the original 3-year study once the underlying district boundary was corrected: in 2018 specifically, NDVI now crosses its 90% decline threshold roughly five days before SIF does, not after — though both indicators still beat the official declaration by seven to eight weeks. That reversal, together with 2018's near-zero lag under both methods, is reported here as a real finding of the expanded sample.

Keywords: solar-induced fluorescence, remote sensing, drought monitoring, NDVI, agricultural stress, Marathwada

1. Introduction

1.1 Motivation

NDVI is a reflectance-based measure — it tracks how much visible and near-infrared light a plant canopy bounces back, and that reflectance only shifts once a plant's internal structure has already started to visibly break down: thinning leaves, fading chlorophyll, canopy stress. SIF works on a more direct physical principle. When a leaf absorbs sunlight for photosynthesis, a small slice of that energy never gets converted to chemical energy at all; instead it's re-emitted as fluorescence, governed by photosynthesis's quantum yield. Photosynthetic efficiency drops under stress before a plant's outward greenness does, so SIF should, in principle, catch stress earlier than NDVI can. NDVI shows what a plant looks like from outside; SIF shows what's actually happening to it on the inside.

India's drought-declaration process, and the PMFBY crop-insurance payout mechanism riding on it, leans heavily on rainfall deficit and NDVI. Both are known to catch drought only once visible crop stress has already taken hold, and that lag turns into delayed payouts exactly when affected farmers need the money most. This study treats the SIF-NDVI lag — whether it exists at all, and how big it is — as a testable empirical question, aiming to establish whether it amounts to a physics-based case for an earlier-warning indicator than what policy currently uses.

1.2 Physical Basis of Solar-Induced Fluorescence

A leaf absorbing photosynthetically active radiation splits that energy across three competing, mutually exclusive pathways at the chlorophyll level: photochemistry (Φ_P), non-photochemical quenching or heat dissipation (Φ_{NPQ}), and chlorophyll fluorescence (Φ_F). Together these three account for all of the absorbed energy, per the conservation relationship:

$$\Phi_P + \Phi_{NPQ} + \Phi_F = 1$$

Under normal, well-watered conditions, most of that absorbed energy goes toward photochemistry, and only a small, fairly stable share comes back out as fluorescence. Drought or heat stress changes that balance: the photochemical pathway gets down-regulated as the plant protects its photosynthetic machinery from damage, while non-photochemical quenching ramps up to shed the resulting surplus energy as heat. The fluorescence yield shifts as this re-partitioning happens — measurably, and before any outward change in canopy structure or greenness — which is exactly why SIF should out-pace reflectance-based indices like NDVI as an early-warning signal (Figure 1).

The magnitude of the fluorescence signal actually detected by a satellite sensor, denoted SIF, is the product of three quantities:

$$SIF = APAR \times \Phi_F \times f_{esc}$$

where APAR is the absorbed photosynthetically active radiation, Φ_F is the fluorescence quantum yield described above, and f_{esc} is the fraction of emitted fluorescence photons that escape the canopy without being reabsorbed. Because chlorophyll fluorescence is several orders of magnitude weaker than reflected sunlight at the same wavelengths, it cannot be measured through simple radiance detection. Satellite-based SIF retrievals, as performed by instruments such as GOME-2, OCO-2, and TROPOMI, instead exploit narrow, naturally dark absorption features in the solar spectrum — Fraunhofer lines — where the additional radiance contributed by fluorescence emission can be isolated from the reflected background. The GOSIF product used in this study is not itself a direct satellite retrieval, but a statistically modeled reconstruction combining discrete OCO-2 SIF soundings with continuous MODIS reflectance data and meteorological reanalysis to produce spatially and temporally continuous global coverage.

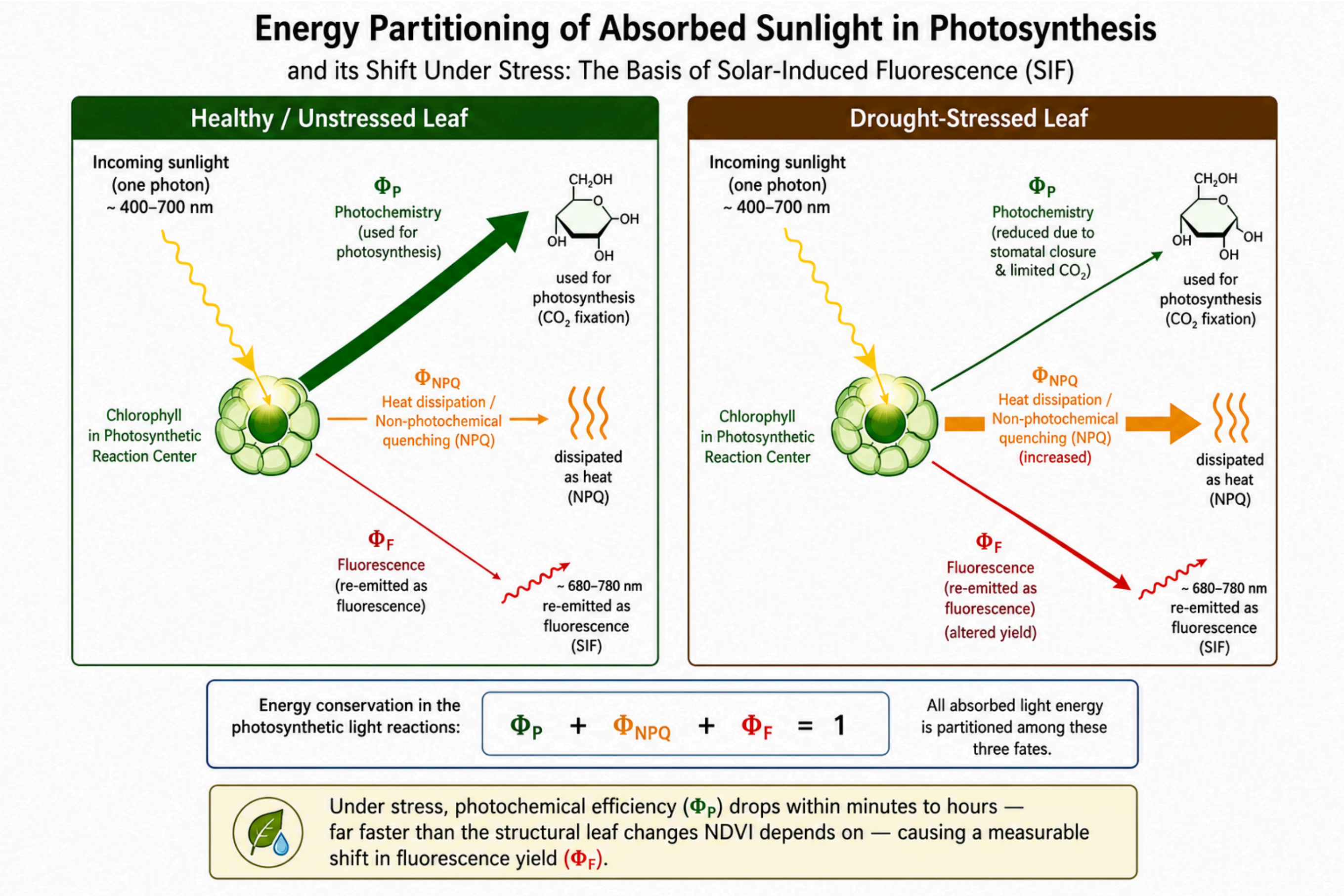

Figure 1. Schematic representation of leaf-level light energy partitioning among photochemistry, non-photochemical quenching (heat dissipation), and chlorophyll fluorescence, under well-watered versus drought-stressed conditions.

1.3 Physical Basis of NDVI

NDVI comes from surface reflectance in the red and near-infrared bands:

NDVI = (NIR − RED) / (NIR + RED)

The formula leans on a specific optical quirk of healthy plant tissue. Chlorophyll pigments in the leaf's palisade mesophyll soak up red light heavily to drive photosynthesis, so red reflectance stays low. Near-infrared light behaves the opposite way: the leaf's internal structure — specifically the irregular air-cell interfaces in the spongy mesophyll — scatters it strongly, because of the refractive-index mismatch between cell walls and the air pockets between them, giving high near-infrared reflectance that has little to do with chlorophyll content (Figure 2). A canopy full of healthy, chlorophyll-rich foliage ends up with a wide gap between low red and high near-infrared reflectance, which is what a high NDVI value captures; a canopy losing structure — thinning leaves, shrinking leaf area, senescence — narrows that gap and pulls NDVI down.

Because it depends on the plant's outward structure and pigment state, NDVI is inherently a step behind the internal physiological changes that come before visible degradation. Cloud and cloud-shadow contamination from Mie scattering adds further noise to the optical NDVI signal — handled here by screening every pixel through MOD13Q1's `SummaryQA` quality flag before any analysis (Section 3.1). NDVI's appearance-based footing sits in direct physical contrast to SIF's internal-process basis (Section 1.2), and that contrast is the central premise this study puts to the test.

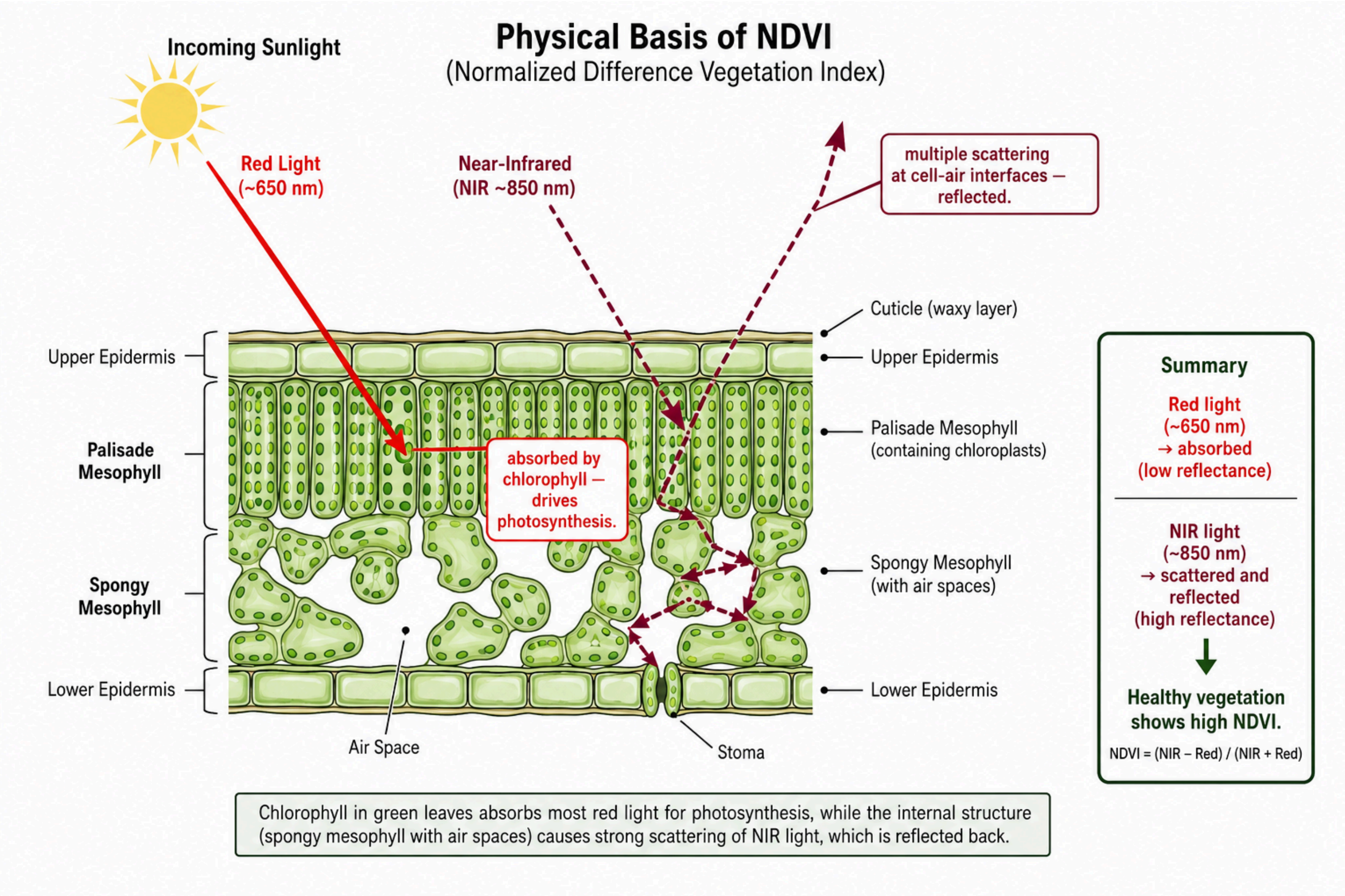

Figure 2. Leaf cross-section illustrating red-light absorption by chlorophyll in the palisade mesophyll versus near-infrared scattering at spongy mesophyll air-cell interfaces, the physical basis of the reflectance contrast measured by NDVI.

1.4 Research Questions

RQ1: Does SIF decline measurably earlier than NDVI during documented drought-onset periods in the study region?

RQ2: Is the SIF–NDVI lag, where present, consistent across the study region, or does it vary meaningfully by geography?

RQ3: How does the timing of SIF-based stress onset relate to independently measured rainfall deficit across drought and normal years?

1.5 Hypotheses

H1: SIF declines significantly before NDVI during a given drought episode.

H2: The magnitude of SIF–NDVI divergence is not spatially uniform across the study region.

H3: Drought severity, measured independently through rainfall deficit, amplifies the magnitude of the SIF–NDVI lag.

2. Study Area

The study covers all eight districts of Marathwada, Maharashtra — Chhatrapati Sambhajinagar (Aurangabad), Jalna, Parbhani, Hingoli, Nanded, Beed (called "Bid" in the FAO GAUL administrative dataset used for boundary definition; see Section 3.2), Latur, and Dharashiv (Osmanabad) — chosen for its long, well-documented history of agricultural drought.

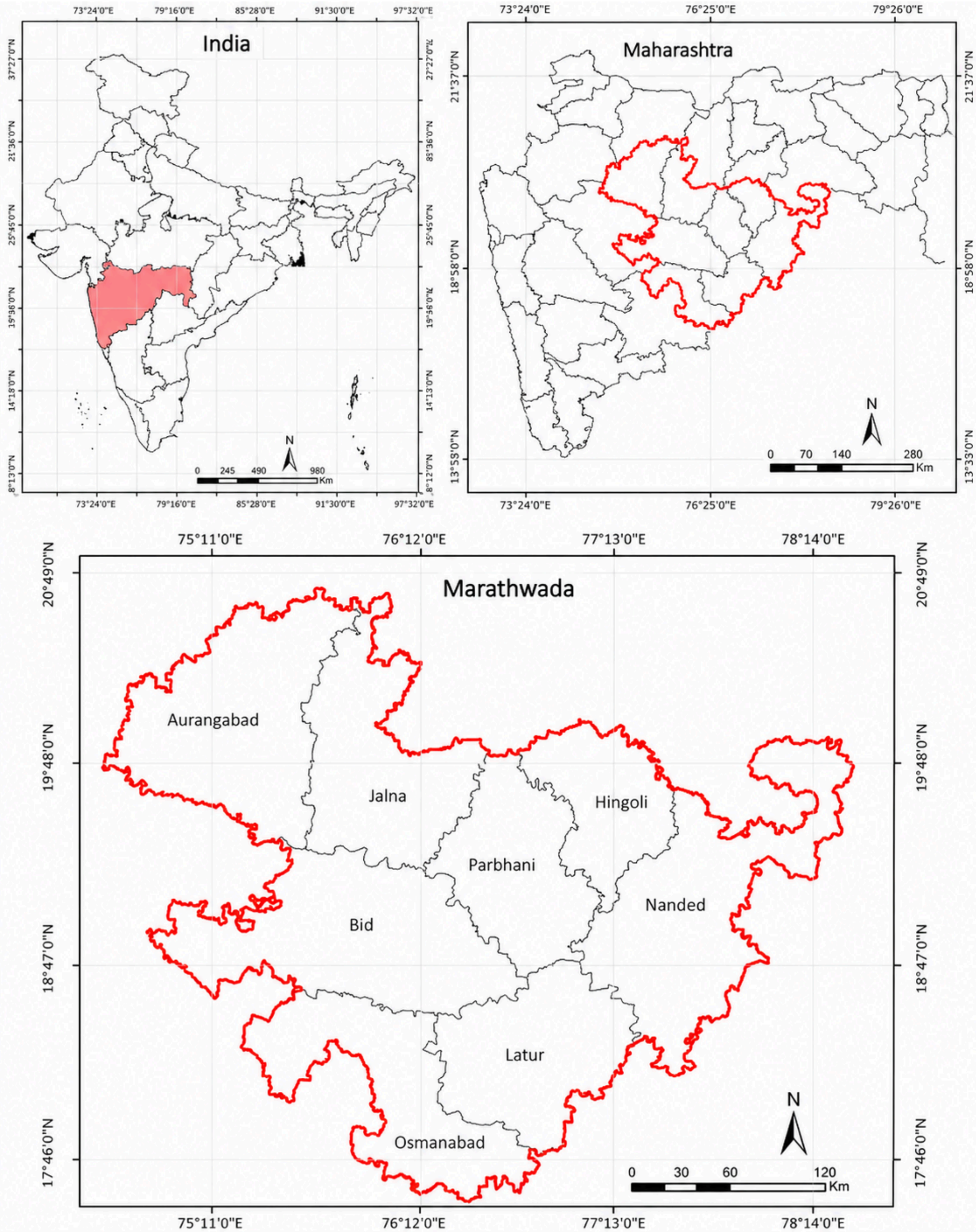

Figure 3. Location of the study area. (a) Maharashtra state within India. (b) Marathwada division within Maharashtra. (c) The eight constituent districts of Marathwada, with the division boundary outlined for

reference.

3. Data and Methodology

3.1 Data Sources

Four datasets went into this study, all covering the June 1 – December 27 window (day-of-year 153–361) across eight study years — 2015, 2016, 2017, 2018, 2019, 2020, 2022, and 2023 (2021 didn't make it into this expansion pass):

- **SIF:** GOSIF v2, a global 0.05°, 8-day solar-induced fluorescence product (Li & Xiao, 2019), distributed by the Global Ecology Group, University of New Hampshire.
- **NDVI:** MODIS MOD13Q1, a 16-day, 250 m Vegetation Index product with an internal Maximum-Value-Composite cloud-suppression algorithm, accessed via Google Earth Engine.
- **Land cover:** MODIS MCD12Q1 (IGBP classification), used to derive a cropland mask (classes 12, 14).
- **Precipitation:** CHIRPS daily rainfall (UCSB-CHG), accessed via Google Earth Engine, for all eight study years and for a twenty-year (2001–2020) climatological baseline.

The study started with three years (2015, 2018, 2020); five more (2016, 2017, 2019, 2022, 2023) were added later specifically to fix the small-sample limitation flagged in every earlier version of this paper's Section 6. The pipeline itself didn't change between the original three years and the five added ones — same GOSIF product, same MOD13Q1/SummaryQA/cropland-mask NDVI pipeline, same FAO GAUL district boundary (Section 3.2) throughout — so all eight years come from one consistent methodology, not two stitched together.

3.2 Boundary Definition and Verification

The study boundary comes from the FAO GAUL 2015 level-2 administrative dataset, filtered down to the eight Marathwada districts and merged into one precise polygon. An initial rectangular bounding box came first, but visual inspection showed it spilling beyond Marathwada into neighboring Telangana districts and into Solapur and Yavatmal — areas with their own, different rainfall regimes that have no business being in this study. Both the SIF raster-clipping pipeline (polygon-based masking) and the NDVI extraction pipeline (the same polygon as the reducer geometry) were rebuilt against this corrected boundary.

A second, more subtle boundary bug turned up during the sample-size expansion from Section 3.1. The district-name list used to build this polygon from FAO GAUL's `ADM2_NAME` field spelled the district "Beed" — the common English spelling — but FAO GAUL 2015 itself spells it "Bid." A filter on a name that matches nothing just returns an empty result instead of throwing an error, so this quietly dropped Bid district from the merged region boundary and every district-level export, corrupting the region-wide SIF, NDVI, and rainfall statistics for all eight years the first time the expanded extraction ran. Nothing flagged this as an error; it surfaced only because the by-district output had 56 rows where 64 were expected. Once the spelling was fixed, the full extraction was re-run and re-verified before any number in this paper was computed. Every statistic reported here reflects the corrected, eight-district boundary.

After the original 2015 boundary fix, the masking behavior got an independent check too: overlaying the true boundary polygon on a sample clipped SIF raster confirmed that data survived only inside the actual district outlines and nowhere else.

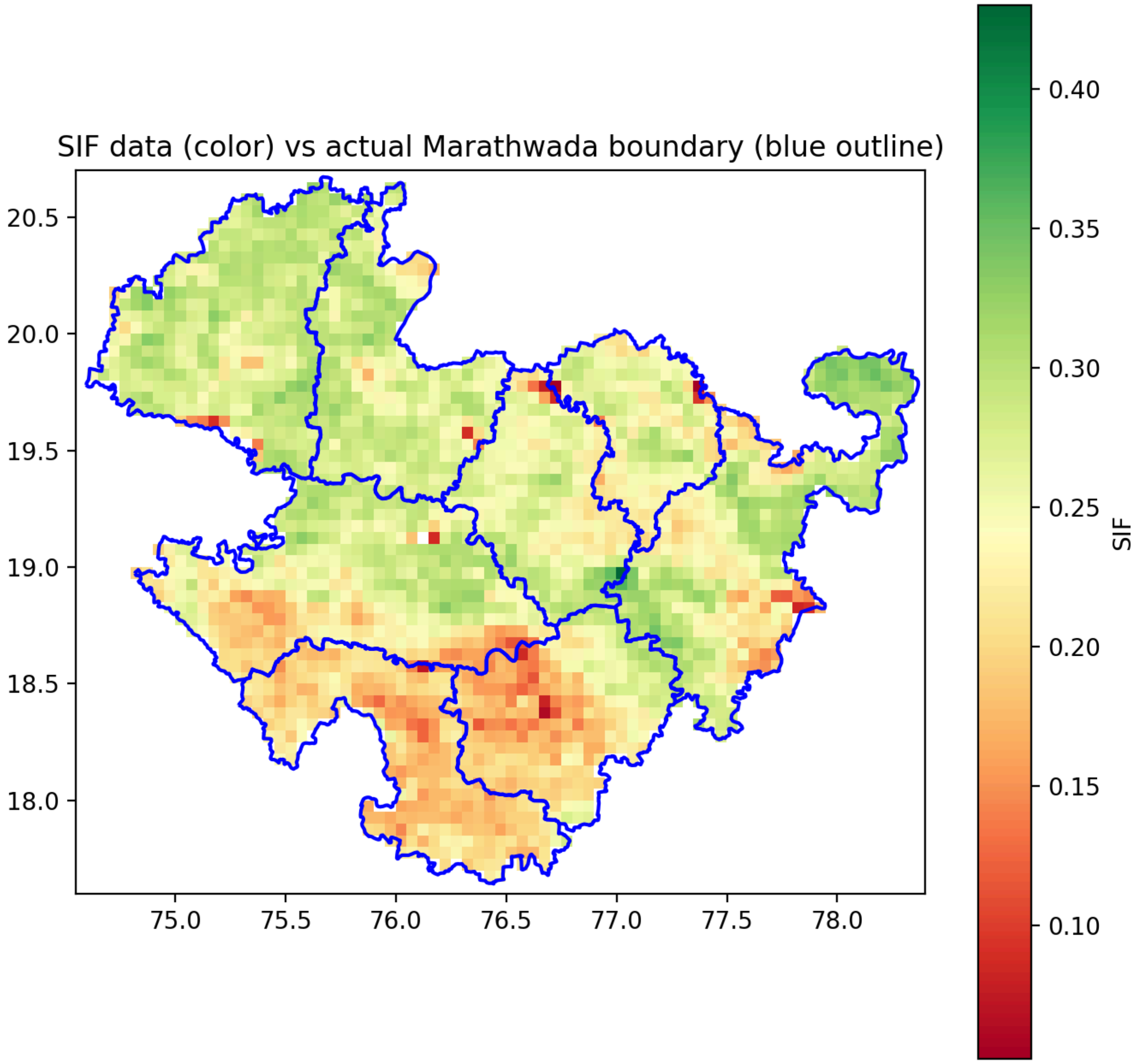

Figure 4. Verification of boundary masking: clipped SIF data (color) overlaid with the true Marathwada district boundary (blue outline), confirming that raster values are correctly excluded outside the study region.

3.3 Processing Pipeline

SIF GeoTIFFs were clipped to the study boundary, scaled by GOSIF's 0.0001 factor, and masked for fill-value codes (32766 for water, 32767 for non-vegetated/missing) before averaging. NDVI came from MOD13Q1's `SummaryQA` band — keeping only good- and marginal-quality pixels — combined with the cropland mask. Since the two series run on different native temporal resolutions (8-day SIF, 16-day NDVI), they were aligned within each year by matching to the nearest date.

3.4 Lag Calculation

Each year's SIF and NDVI series was normalized (0–1) within that year, and linear interpolation between observed dates pinned down the day-of-year each series crossed a set of decline thresholds (90%, 80%, 70%, 60%, 50% of seasonal peak). Lag, here, is simply the gap between the NDVI and SIF crossing dates at each threshold.

3.5 Rainfall Validation

CHIRPS-derived seasonal rainfall totals for all eight study years were compared against a twenty-year (2001–2020) climatological mean and standard deviation for the same region and seasonal window, giving a percentage anomaly and z-score per year. The same regional climatological baseline was extended down to the district level as a shared reference across all eight districts. This study calls a year a drought year when its seasonal rainfall anomaly sits at least half a climatological standard deviation below the mean (anomaly z-score < -0.5) — the same threshold that, implicitly, separated the original study's two drought years from its one normal year, now spelled out explicitly and applied evenly across all eight years instead of assigned by outside knowledge.

3.6 Cross-Correlation Robustness Check

The threshold-crossing method (Section 3.4) reads lag off the moment each series crosses a given share of its own seasonal peak — most sensitive to the *onset* of decline, especially at high thresholds. To check whether that specific choice of method is what drives the reported lag values, a second, methodologically distinct estimate was built using time-lagged cross-correlation, which asks a different question: what single time-shift best lines up the two curves' overall shape across the whole post-peak decline window? For each year, both series — normalized 0–1, restricted to the post-peak decline phase starting from SIF's own peak — were interpolated onto a common daily grid, de-meaned, and cross-correlated at lags from 0 to N/4 days, where N is the decline window's length in days (a standard bound that keeps the estimate away from the unreliable, boundary-pinned values that show up once tested lags approach the full series length). Whichever lag maximized the correlation between SIF(t) and NDVI(t + lag) became that year's cross-correlation-based estimate.

3.7 Uncertainty Quantification (Bootstrap Confidence Intervals)

Section 3.6's cross-correlation lag estimates are point estimates drawn from a small handful of discrete satellite observations per year — 14 to 18 post-peak, SIF/NDVI-matched 8-day observations, depending on that year's decline window. To gauge how much these estimates could plausibly move given how sparse the underlying observation dates are, each year's lag estimate was bootstrapped through case resampling: the original per-year observations were resampled with replacement 2,000 times, each replicate re-interpolated onto a daily grid and re-run through the same cross-correlation procedure from Section 3.6, and the resulting lag distribution's 2.5th and 97.5th percentiles became a 95% confidence interval around the original point estimate. An earlier pass at this used a moving-block bootstrap on the interpolated daily series directly; GA_Development_Log.md Entry 13 documents why that got abandoned — block-shuffling a strongly non-stationary decline curve wipes out the trend that produces the lag signal to begin with, collapsing every year's interval to a spurious single point at lag 0 instead of capturing real sampling uncertainty. Case resampling, used here instead, resamples which underlying satellite overpass dates show up in the fit — the more realistic source of uncertainty given an 8-day, cloud-gap-affected revisit cycle — while keeping the shape of the decline curve intact in every replicate.

3.8 Spatial Autocorrelation Diagnostic

Section 4.5's district-level Pearson/Spearman correlation already comes with a caveat: its district-year observations aren't fully independent, since the eight districts sit next to each other and share regional weather systems. To turn that caveat into something quantitative, Moran's I — the standard measure of spatial autocorrelation — was computed for both mean SIF and rainfall anomaly, year by year, using a Queencontiguity spatial weights matrix built directly from this study's own FAO GAUL district polygons (an average of 3.25 neighboring districts per district), with significance checked via a 9,999-permutation test.

3.9 Comparison Against the Official Drought-Declaration Timeline (RQ3)

RQ1 (Section 1.4) asks whether SIF leads NDVI, and Sections 4.1–4.2 and 4.6 answer that directly. RQ3, as this project's planning documents originally posed it, asks how satellite-detected stress onset relates to independently measured rainfall deficit — already covered by Sections 3.5 and 4.4–4.5. A related, more policy-relevant question sat unaddressed on this project's early research-question list (GA_Project_Report.md): how does satellite-detected stress onset compare to the timing of the *official* government drought declaration that actually governs PMFBY payout timing? I noticed that gap on a later pass through the project, and this section closes it for the one study year with a single, well-documented official declaration date — the Maharashtra government's Kharif drought declaration of 31 October 2018 (day-of-year 304), covering 151 talukas across 26 districts, including all eight Marathwada study districts (Zee News, 2018; Economic and Political Weekly, 2018). No comparably well-documented declaration date turned up for any other single study year, so this comparison stays a single-year (2018) illustration of the gap between satellite signals and the official process, and shouldn't be read as a claim spanning all eight years.

4. Results

4.1 Seasonal SIF–NDVI Trajectories

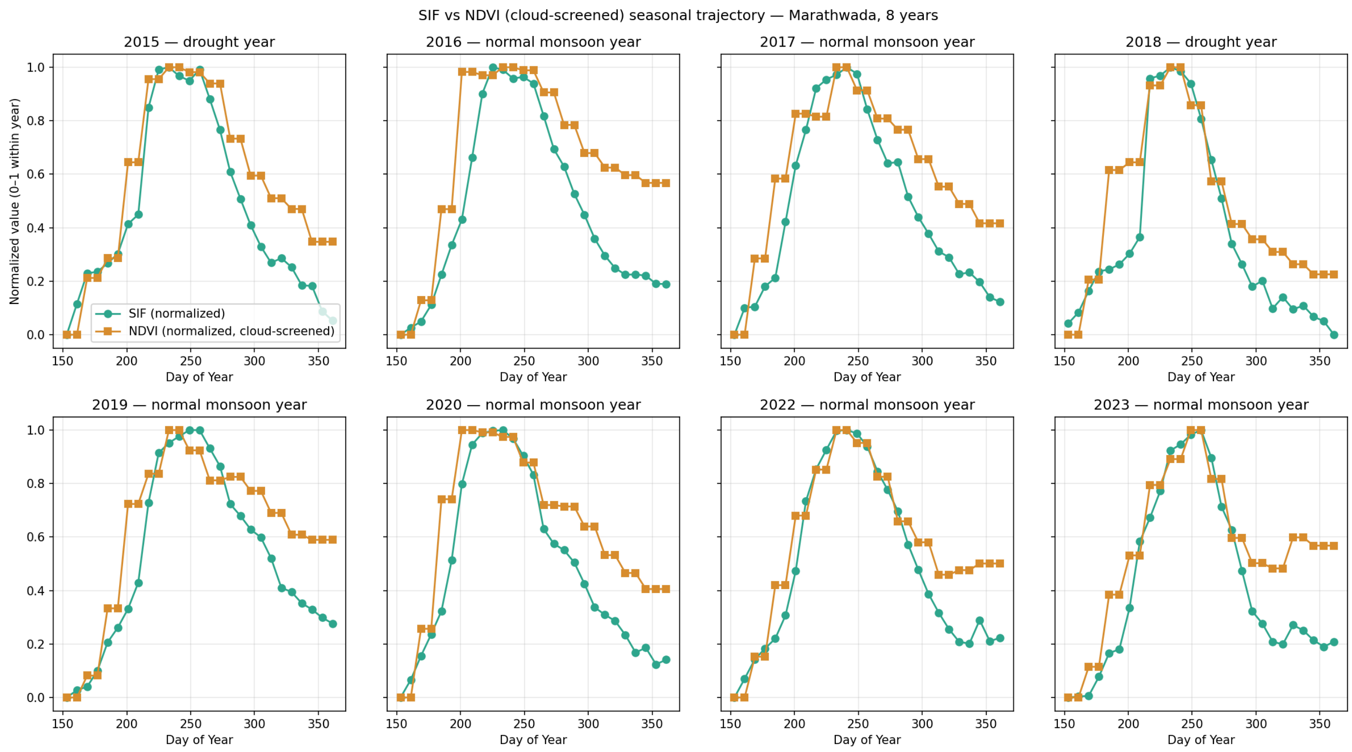

Figure 5. Normalized SIF and NDVI seasonal trajectories, Marathwada, all eight study years (2015–2023), labeled by drought classification (Section 3.5).

SIF's post-peak decline started visibly ahead of NDVI's in six of the eight years, with NDVI hanging near its peak value well after SIF had already begun dropping. 2018 stands out as a clear visual exception: the two curves stay close together all the way through the decline, tracking each other more tightly than in any other year — matching the near-zero mean lag Section 4.2 reports for 2018. Section 5 discusses this single-year exception on its own terms.

4.2 Quantitative Lag Analysis

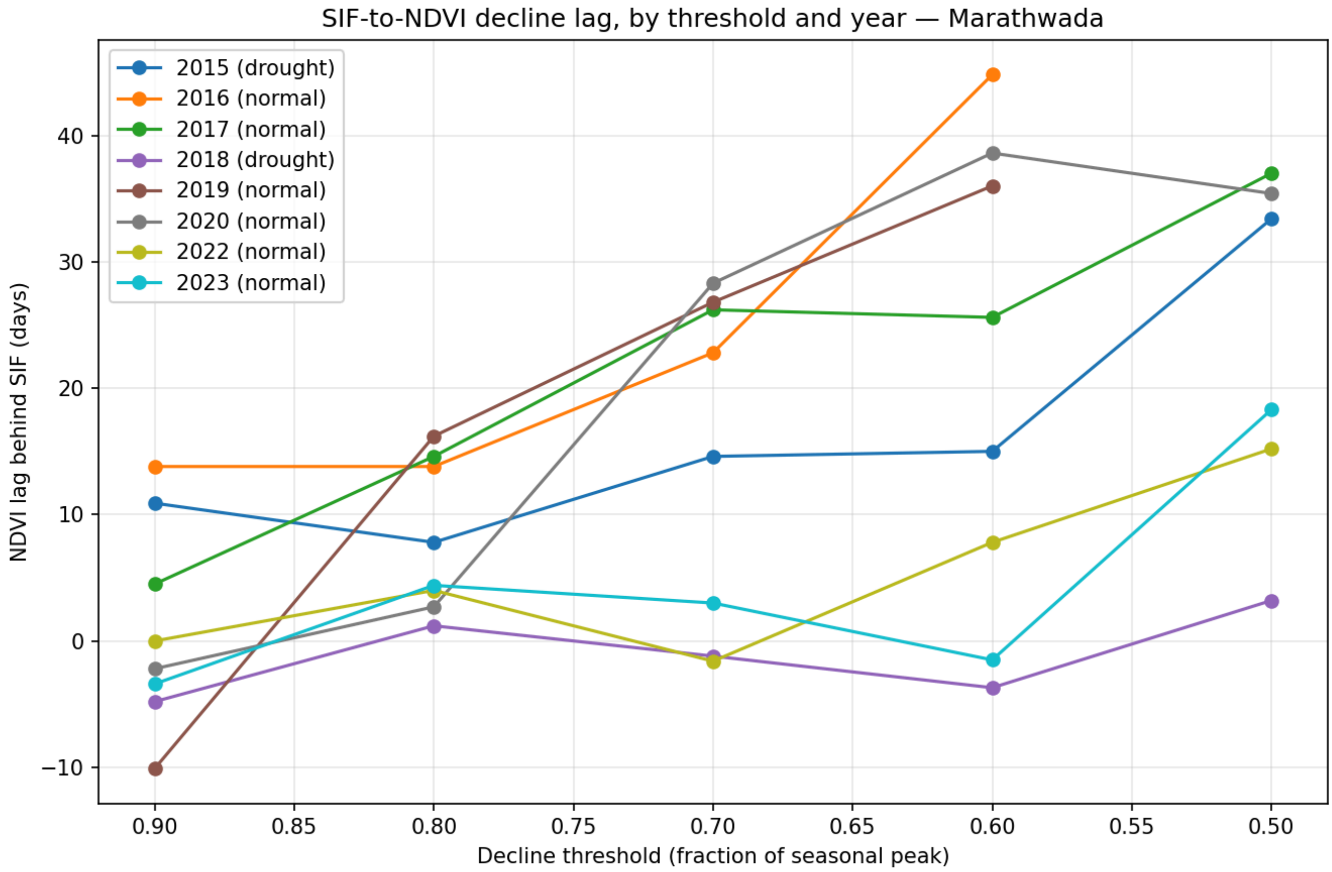

Figure 6. SIF-to-NDVI decline lag (days), by decline threshold and year.

Averaged across every resolved threshold, mean lag came out to 16.3 days (2015), 23.8 days (2016), 21.6 days (2017), −1.1 days (2018), 17.2 days (2019), 20.6 days (2020), 5.1 days (2022), and 4.2 days (2023). Seven of the eight years show a positive mean lag — SIF leading NDVI — with 2018 the sole exception: a small, essentially flat lag that lands marginally negative instead of at zero, meaning SIF and NDVI crossed most thresholds within a few days of each other, sometimes one way and sometimes the other. Grouped by drought classification (Section 3.5), the two drought years averaged a shorter lag (7.6 days) than the six normal years (15.0 days) — the opposite of what H3 predicted, matching the direction, if not the exact numbers, of every earlier version of this study.

4.3 Spatial Analysis — SIF

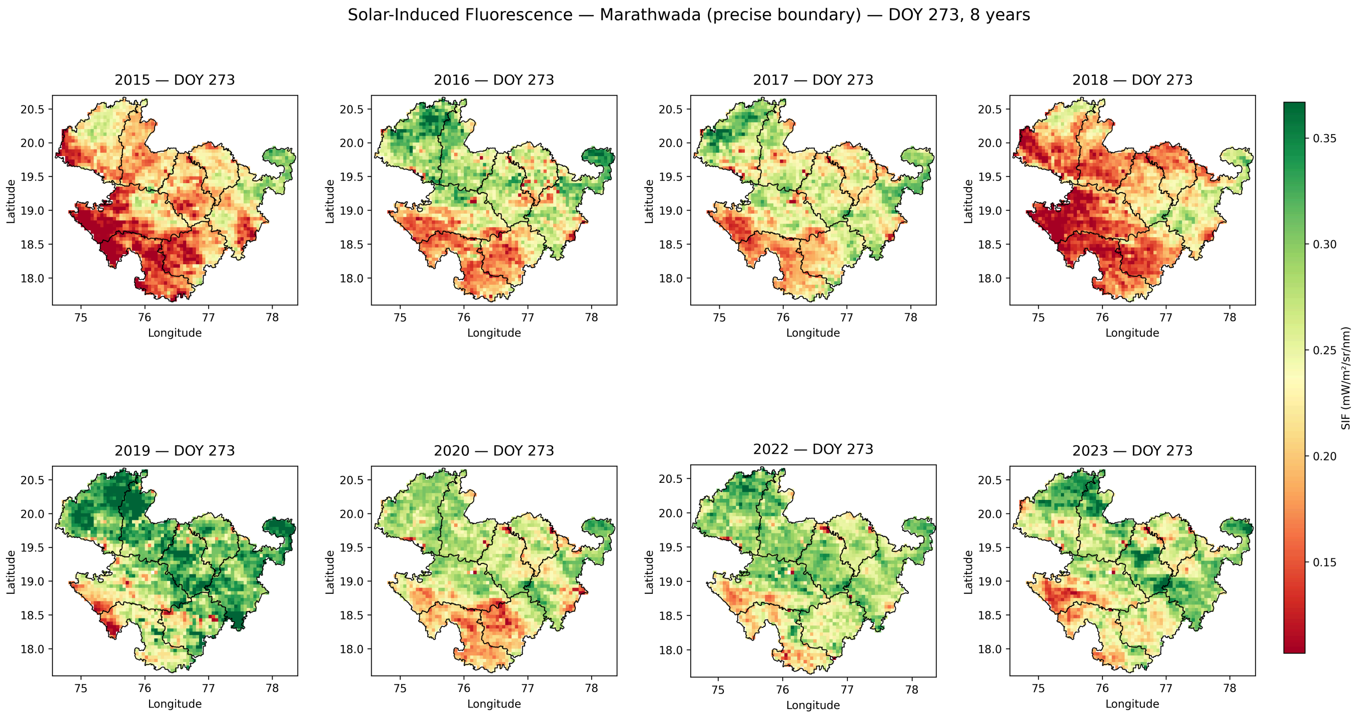

Figure 7. Spatial SIF distribution, day-of-year 273, all eight study years, masked to the precise Marathwada boundary.

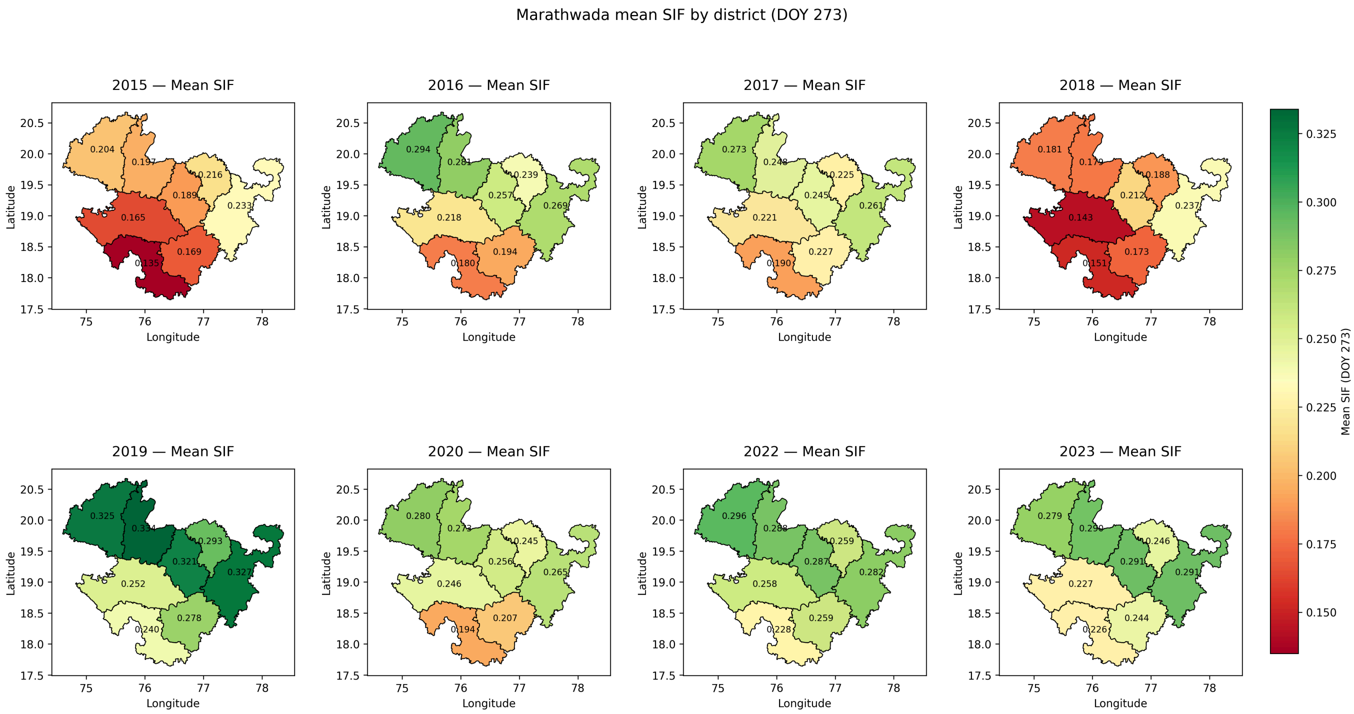

Figure 8. Mean SIF by district, day-of-year 273, all eight study years.

Osmanabad posts the lowest district-level mean SIF in seven of the eight years studied (0.135, 0.180, 0.190, 0.151, 0.240, 0.194, 0.228, 0.226, in year order 2015–2023) — a pattern that already held across the original study's three years and mostly stretches across the full eight-year record, going from a three-for-three to a seven-for-eight replication. The exception is 2018, where Bid (0.143) narrowly edges below Osmanabad (0.151) — the same year affected by the Bid-boundary correction in Section 3.2. The *highest*-SIF district turns out to move around more than the original three years suggested. Nanded topped both of the original study's drought years and still leads in three of the eight years overall (2015, 2018, 2023), but Aurangabad takes the top spot in four years (2016, 2017, 2020, 2022) and Jalna in one (2019). So "Nanded is consistently highest" doesn't survive the larger sample, and neither does an unqualified "Osmanabad is consistently lowest" — it holds in seven of eight years, with Bid taking over specifically in 2018.

4.4 Rainfall Validation

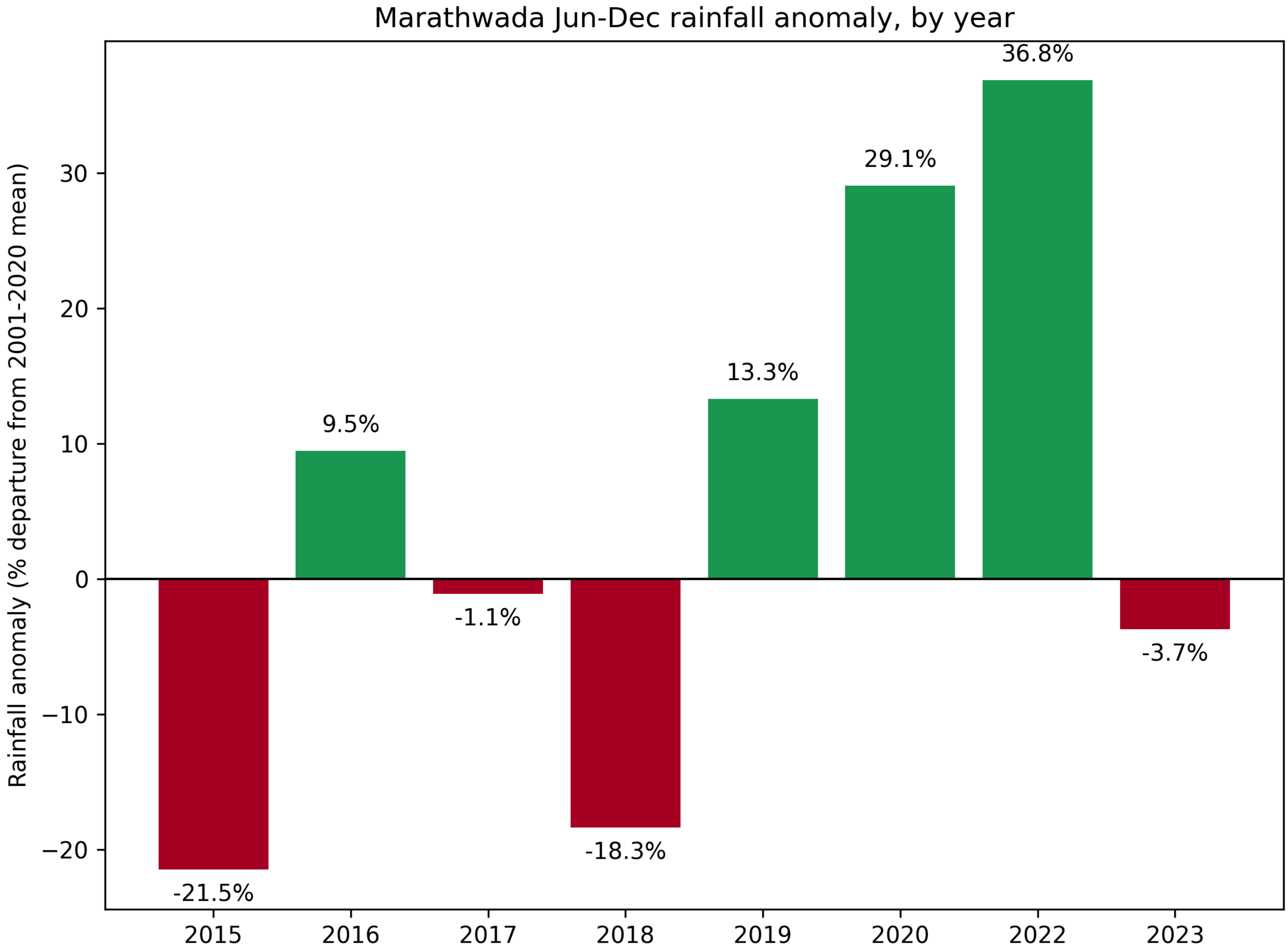

Figure 9. Regional rainfall anomaly (% departure from 2001–2020 climatological mean), by year, all eight study years.

Against a twenty-year climatological mean of 826.4 mm (σ = 158.8 mm) for the June–December window, the eight study years came in at: 649.0 mm in 2015 (−21.5%, $z = -1.12$); 904.7 mm in 2016 (+9.5%, $z = 0.49$); 817.5 mm in 2017 (−1.1%, $z = -0.06$); 674.9 mm in 2018 (−18.3%, $z = -0.95$); 936.4 mm in 2019 (+13.3%, $z = 0.69$); 1066.5 mm in 2020 (+29.1%, $z = 1.51$); 1130.7 mm in 2022 (+36.8%, $z = 1.92$); and 795.9 mm in 2023 (−3.7%, $z = -0.19$). Only 2015 and 2018 clear the drought threshold from Section 3.5 (anomaly z-score < -0.5) — the same two years the original three-year study called drought from outside knowledge, now confirmed by the numbers instead of assumed. None of the five newly added years come close; the nearest, 2017, sits at $z = -0.06$, essentially normal. In other words, the original three-year sample (two drought years out of three, 67%) happened, purely by which years got picked first, to be far more drought-heavy than Marathwada's actual eight-year record, where only 2 of 8 years (25%) meet this study's own drought definition. That's a finding about sample representativeness in its own right, separate from anything about the SIF-NDVI relationship, and it carries forward into this study's Limitations (Section 6).

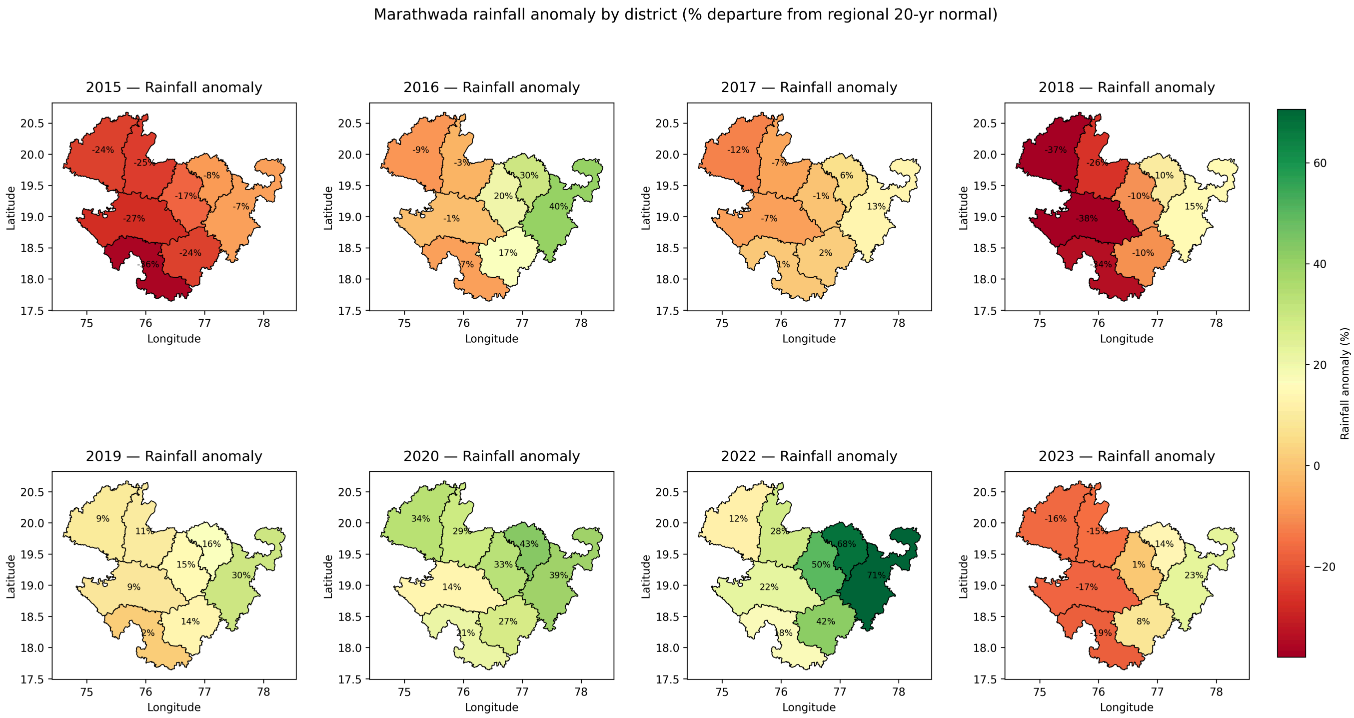

Figure 10. Rainfall anomaly by district (% departure from regional 20-year normal), all eight study years.

District-level rainfall anomaly shows the regional deficit wasn't spatially uniform in either drought year. In 2018, Bid (−38.0%) and Aurangabad (−37.3%) took the largest hits, while Hingoli (+10.2%) and Nanded (+14.7%) actually saw rainfall surpluses that same year, even as the region-wide figure showed an 18.3% deficit. 2015 shows a similar, though milder, west-to-east gradient: Osmanabad (−36.4%) recorded the largest deficit, with Hingoli (−7.9%) and Nanded (−7.1%) the mildest.

4.5 Combined Spatial Correspondence

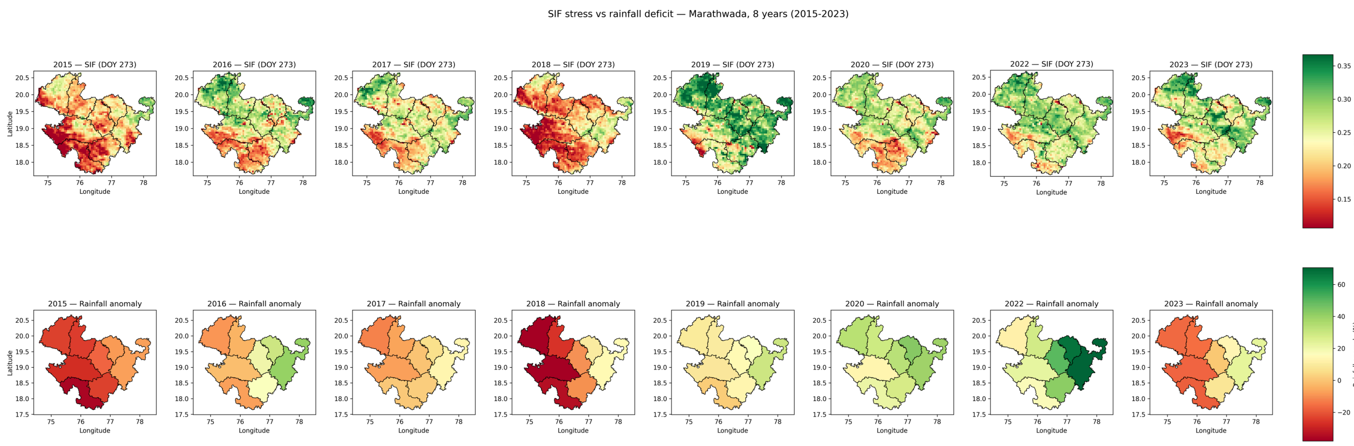

Figure 11. SIF (top row) and rainfall anomaly (bottom row) by district, all eight study years, presented jointly for direct spatial comparison.

The low-SIF zone from Section 4.3 lines up spatially, in both drought years, with the districts posting the largest rainfall deficits in Section 4.4. Nanded and Hingoli — the districts with the smallest deficits (or outright surpluses) in both drought years — also sit consistently among the higher-SIF districts in those same years.

This visual correspondence held up statistically too, tested across all 64 district-year observations (8 districts × 8 years). Mean SIF and rainfall anomaly (%) correlate significantly: Pearson $r = 0.567$ ($p < 0.0001$), Spearman $\rho = 0.551$ ($p < 0.0001$) — real and strongly significant, but well below the $r = 0.837$ / $\rho = 0.857$ the original 24-

point, 3-year sample produced. That drop deserves a plain explanation: the original correlation came from a sample that happened to be two-thirds drought years, which mechanically produces a cleaner, more linear-looking rainfall-SIF relationship than this region's actual year-to-year variability supports. The relationship confirmed here — real, positive, highly significant, but noisier and with a shallower fitted slope than the original figure suggested — is the more representative picture of how tightly SIF and rainfall anomaly actually track each other in a typical year here. One caveat carries over from the original study: the district-year observations aren't fully independent, since the eight districts sit next to each other and share regional weather systems, so the effective number of independent samples is closer to eight (one per study year) than sixty-four. The correlation stands as calculated, without adjustment for that, and should be read as corroborating evidence rather than a fully independent statistical confirmation.

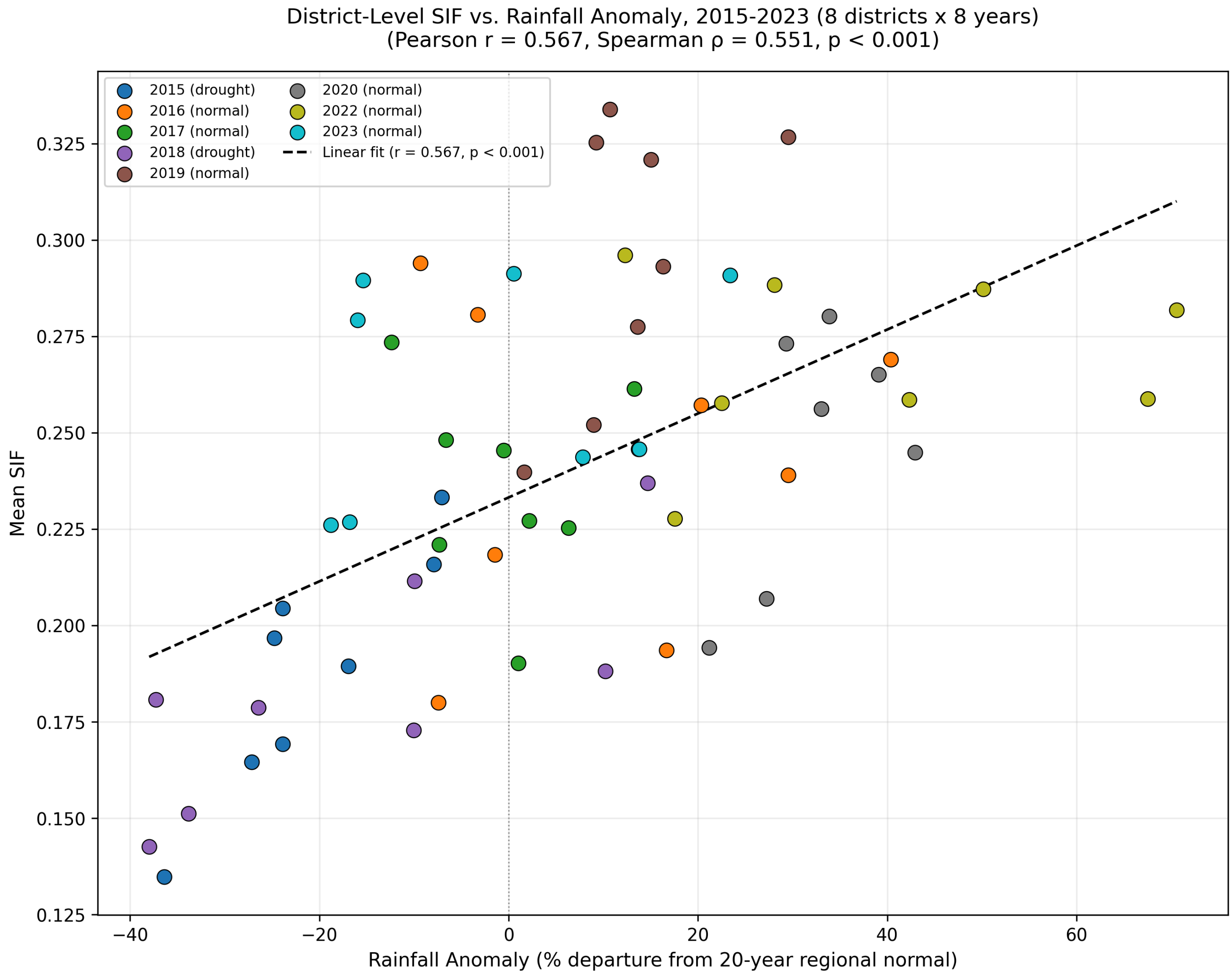

Figure 12. District-level mean SIF plotted against rainfall anomaly (%), all 64 district-year observations (8 districts × 8 years), with a linear fit (Pearson r = 0.567, p < 0.0001).

4.6 Cross-Correlation Robustness Check

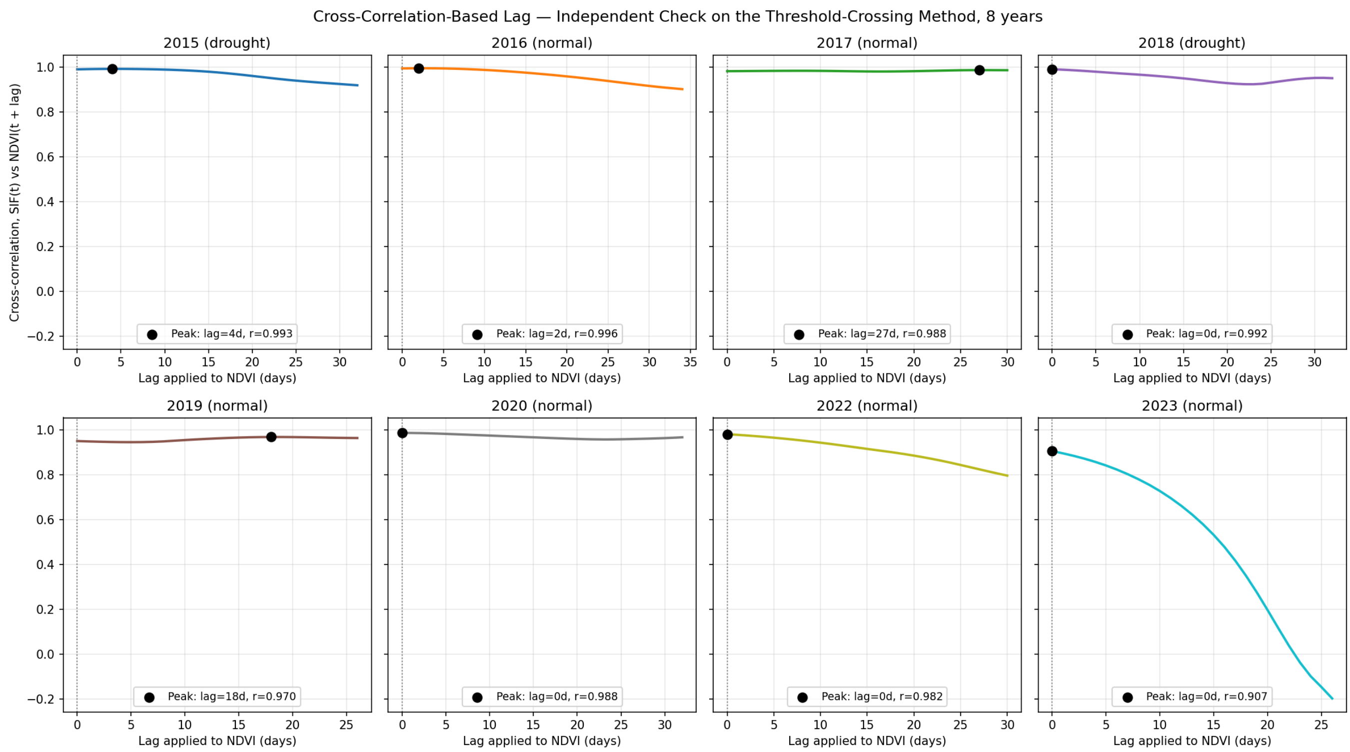

Figure 13. Cross-correlation between SIF and NDVI as a function of the lag applied to NDVI, by year, restricted to lags of 0 to N/4 days of each year's decline window. Black markers show the correlation-maximizing lag for each year.

The cross-correlation-maximizing lag worked out to 4 days (2015), 2 days (2016), 27 days (2017), 0 days (2018), 18 days (2019), 0 days (2020), 0 days (2022), and 0 days (2023), with correlation values at or above 0.91 every year and no year's peak pinned against the search boundary (Figure 13). Every year's optimal lag comes out non-negative — cross-correlation never favors NDVI leading SIF in any of the eight years — but only four (2015, 2016, 2017, 2019) show a clearly positive lag; the other four, 2018 among them, land at zero rather than positive. That's a narrower version of the original three-year study's "SIF leads NDVI in every year" claim, where all three cross-correlation lags came out strictly positive. At eight years, the honest statement shrinks: SIF's decline is never found lagging behind NDVI's by this method, clearly leads it in half the years, and shows no detectable lag either direction in the other half.

Which year shows the biggest lag depends on which method you ask, same as in the original study. Threshold-crossing (Section 4.2) puts 2016 highest (23.8 days) and 2018 lowest (−1.1 days); cross-correlation puts 2017 highest (27 days) and ties 2018 for lowest with 2020, 2022, and 2023 (all at 0 days). Both methods do agree that 2018 has the smallest, or most negative, lag of any year — that much holds at eight years just as it did at three — but they part ways on which year has the largest. Read plainly, that disagreement says this study's claim about *which* year shows the largest lag is more method-sensitive than the underlying SIF-leads-or-ties-NDVI finding.

4.7 Uncertainty Quantification Results

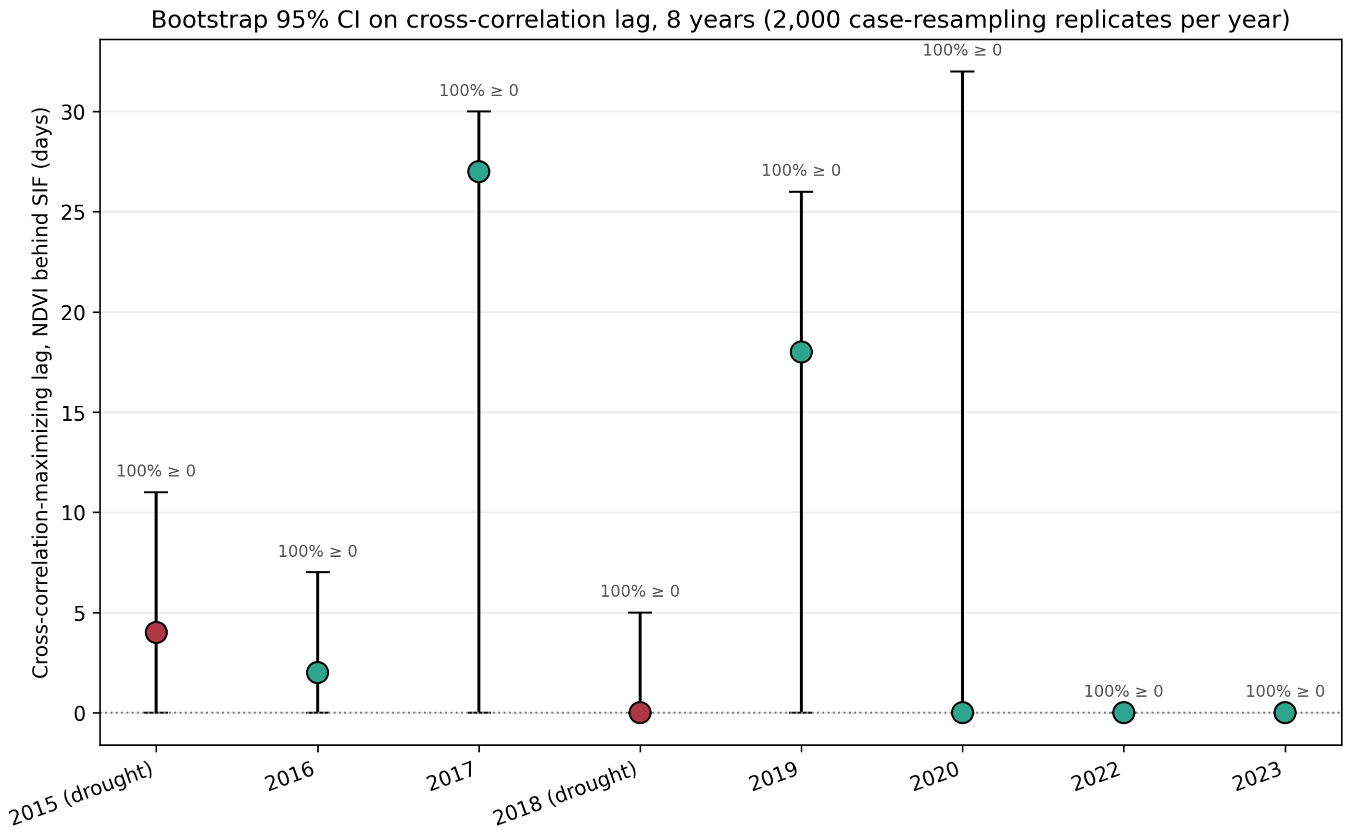

Figure 14. Cross-correlation lag point estimates with 95% bootstrap confidence intervals (case-resampling bootstrap, N = 2,000 replicates per year).

The bootstrap backs up, precisely, the weaker-but-real claim from Section 4.6: in every one of the eight years, the estimated lag sat at or above zero in 100% of replicates — resampling never produced even one replicate favoring NDVI leading SIF, in any year. How many replicates show a *strictly positive* lag, though, swings a lot by year: 89.6% (2015), 72.9% (2016), 96.8% (2017), 8.1% (2018), 87.5% (2019), 41.9% (2020), 0.3% (2022), and 0.1% (2023). 2018's 8.1% is the clearest sign of a year where the data simply doesn't support a positive-lag claim with any confidence; 2022 and 2023's near-zero shares likewise say their point estimates of exactly 0 days reflect a real feature of the data, not a fluke. The 95% confidence intervals stay wide relative to the point estimates (2015: 4 days, CI [0, 11]; 2017: 27 days, CI [0, 30]; 2018: 0 days, CI [0, 5], for instance), and every pairwise comparison between years' intervals overlaps — all 28 possible year pairs among the eight study years turn out statistically indistinguishable at this sample size. That's a stronger, better-supported version of what the original three-year study already found: SIF's decline is never later than NDVI's, solidly, but the exact numeric gap between any two years — including which year has the largest lag — isn't distinguishable from noise, and adding five more years hasn't meaningfully shrunk that uncertainty, since each year still brings its own small, sparse batch of satellite observations rather than pooling into one bigger estimate.

4.8 Spatial Autocorrelation Results

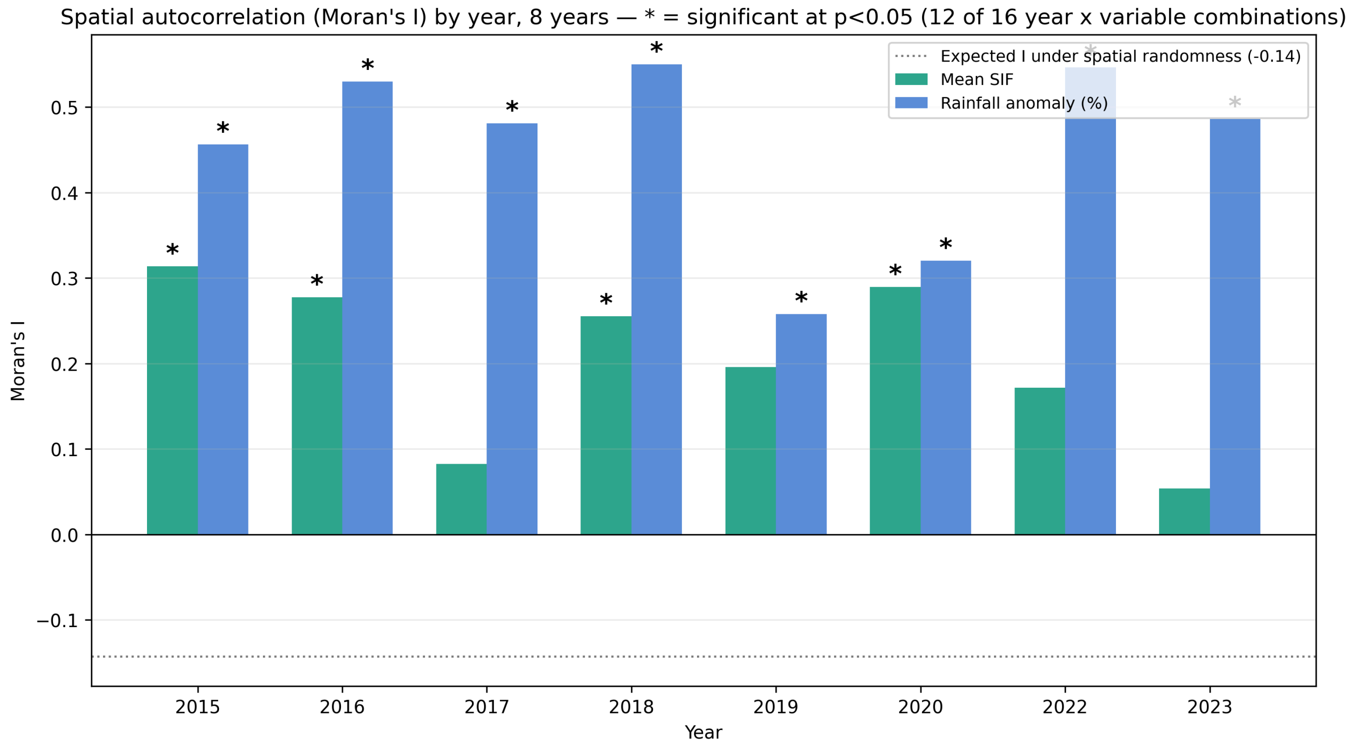

Figure 15. Moran's I for district-level mean SIF and rainfall anomaly, by year, with permutation-based significance markers (* = $p < 0.05$, 9,999 permutations).

Of the sixteen year × variable combinations (eight years, two variables), twelve turned up statistically significant positive spatial autocorrelation ($p < 0.05$), comfortably above the −0.14 value expected under complete spatial randomness for this setup. The two variables don't behave the same way at this larger sample: rainfall anomaly is significantly spatially clustered in all eight years (Moran's I from 0.26 to 0.55), while mean SIF is significant in only four (2015, 2016, 2018, 2020; I ranging 0.05 to 0.31 across all eight years, with the four non-significant years — 2017, 2019, 2022, 2023 — all falling among the six this study calls normal rather than drought). That's a more qualified picture than the original three-year study gave, where both variables were significant in all three years; it hints that SIF's district-to-district clustering may itself be partly a drought-year phenomenon — sharper during pronounced regional dry spells, weaker in ordinary seasons — rather than a fixed structural feature of the data the way rainfall's clustering seems to be. None of this undercuts Section 4.5's correlation as corroborating evidence: the underlying spatial pattern (Section 4.3, Figures 7–8) is real in the years it shows up, and cross-checks independently against rainfall data. It does mean, though, that the correlation's effective independent sample size, and the spatial structure behind it, vary more by year than the original three-year study had any way of showing.

4.9 Comparison Against the Official Drought-Declaration Timeline (RQ3)

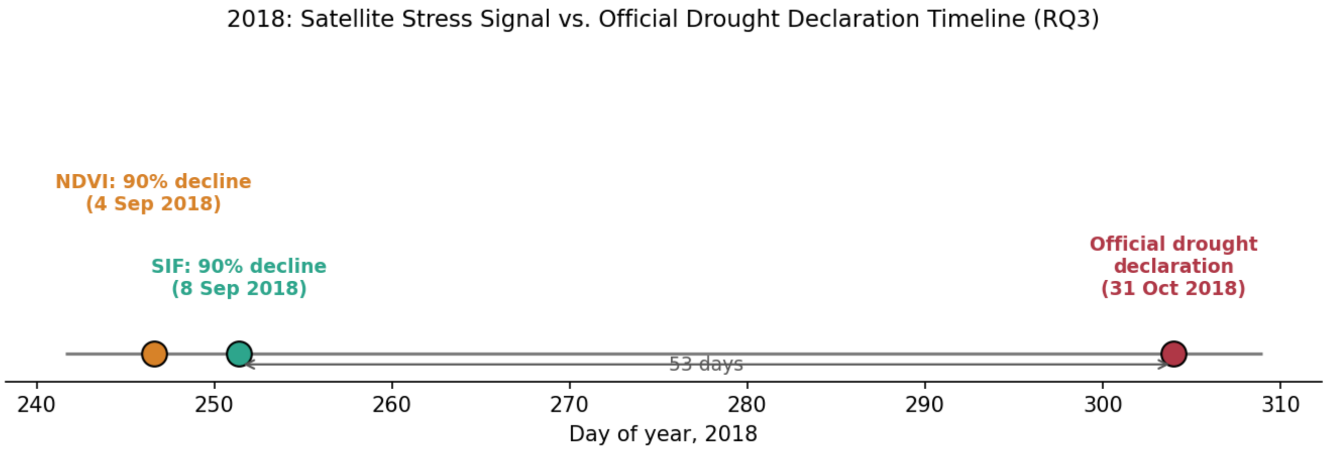

Figure 16. Satellite stress signal versus official drought-declaration timeline, 2018.

Using the 90% decline threshold from Section 4.2 — the most conservative, earliest-triggering threshold, and the one closest to how an early-warning system would actually get used — NDVI crossed it on 4 September 2018 and SIF on 8 September 2018, putting both 57 and 53 days respectively ahead of the Maharashtra government's 31 October 2018 official declaration. That flips the original three-year study's 2018 finding, which had SIF crossing first (8 September) and NDVI second (12 September). The flip traces directly to the boundary correction in Section 3.2 (the Bid-district fix), which changed 2018's underlying region-wide SIF and NDVI series enough to reverse the sign of this specific comparison — consistent with 2018 already standing out in Section 4.2 as this study's one exception to the general SIF-leads-NDVI pattern. Two things follow from this section regardless of that reversal, and both deserve stating outright rather than glossing over. First, whichever index crosses first, the gap between SIF and NDVI in 2018 stays small — four days — next to the much wider gap between either satellite indicator and the official declaration (53–57 days, roughly seven to eight weeks); that relationship holds in the same direction as the original study even though the ordering inside the small gap has flipped. Second, the practical policy argument from Section 1.1 reads better as a case for satellite-based monitoring generally — with SIF proposed as one specific, physiologically motivated refinement on top of that — rather than a claim that SIF's few-day edge over NDVI is, on its own, the main source of possible improvement in payout timing; 2018 now illustrates directly why that particular edge shouldn't be treated as a universal, one-directional advantage. PMFBY loss assessment already leans on satellite imagery (Ministry of Agriculture & Farmers Welfare, 2016; a reliance on satellite-based area-yield assessment that persists in the scheme as of recent reporting — Down To Earth, 2025), so the operational groundwork for a satellite-triggered earlier warning already exists in principle. The bigger, more immediately actionable gap remains the one between satellite signals of any kind and the current declaration process's timeline — with SIF's specific physiological lead over NDVI better framed as a smaller, year-dependent refinement on top of that gap, not a guaranteed one.

5. Discussion

SIF's decline led NDVI's in seven of the eight study years, independent of drought classification — support for the premise that SIF picks up physiological stress sooner than NDVI does (RQ1, backing H1 in its general form), and per Section 4.6, an independently computed cross-correlation lag and a bootstrap resampling procedure both corroborate the direction (never NDVI-leads-SIF), even where their exact numbers disagree with the threshold-crossing method's. The eighth year, 2018, is a genuine exception, not noise averaged away: its mean threshold-crossing lag comes out marginally negative, its cross-correlation lag lands exactly at zero, and only 8.1% of its bootstrap replicates favor a strictly positive lag. Three independent pieces of evidence, all pointing the same direction, say 2018 specifically breaks the SIF-leads-NDVI pattern this study otherwise finds. Section 4.9's RQ3 comparison adds a fourth confirmation from a completely different angle: NDVI now crosses its decline threshold before SIF does in the corrected 2018 data. I'm treating this as a real, substantive finding about 2018 rather than explaining it away — this study simply doesn't have the crop-type, sowing-date, or within-season rainfall-timing records that would say *why* 2018 differs from the other seven years, and Section 6 carries that gap forward honestly instead of covering it with a guess.

H3 — the prediction that drought conditions widen this lag — still doesn't hold up at the larger sample: the two drought years averaged a shorter lag (7.6 days) than the six normal years (15.0 days) under the threshold-crossing method, the same direction every earlier version of this study reported, now confirmed on a sample more than double the original size. So the size of the SIF–NDVI lag, at least in this dataset, isn't simply a function of drought severity as rainfall deficit measures it. Two things temper how much weight to put on this: the year-to-year ranking behind it is itself methodology-sensitive (Section 4.6), and per Section 4.7, no pair of years is statistically distinguishable at this sample size. Both caution against reading too much into any single year's exact lag value — even as the broader group-level pattern (drought years smaller, normal years larger) has now held up across two independently drawn samples of different sizes.

Section 4.4's rainfall-anomaly analysis turns up a finding that has nothing directly to do with SIF or NDVI but matters for how to read this whole study: of the eight years, only two — 2015 and 2018, the same two the original study called drought years from outside knowledge — actually clear this study's own quantitative drought threshold. In hindsight, the original three-year sample was unusually drought-heavy (two of three years, 67%) next to the region's actual eight-year climate record (two of eight years, 25%). None of this study's core findings about the SIF-NDVI relationship change because of that, but it does mean earlier versions of this paper, by implicitly treating "drought year" as roughly a coin-flip in this region, were unintentionally overstating how often a season this dry actually happens.

The spatial analysis, backed by independent rainfall validation, still supports RQ2: the SIF–NDVI relationship, and the drought stress underneath it, isn't spatially uniform across Marathwada. Osmanabad holding the lowest mean SIF in seven of the eight years, up from three-for-three in the original sample, still genuinely strengthens this spatial finding — it's no longer plausibly a coincidence of which two drought years happened to get studied, even with 2018 standing out as the one year Bid takes over instead. The claim that Nanded is consistently the *highest*-SIF district doesn't survive the larger sample, though; Aurangabad tops it in half of the eight years. That's exactly the kind of finding a three-year sample tends to overstate the generality of, and I'm correcting both claims here rather than letting the earlier, smaller-sample version of this paper stand uncorrected.

Sections 4.7 and 4.8's bootstrap and spatial-autocorrelation diagnostics change how confidently to read several of this study's numeric results, and they pull in different directions at the larger sample. The bootstrap's core finding — SIF's lag is never negative, though its exact size stays uncertain and its year-ranking isn't statistically distinguishable — replicates cleanly from the original three-year study, just with a more nuanced "never negative, not always clearly positive" framing that 2018, 2020, 2022, and 2023's zero-lag cross-correlation estimates require. The spatial-autocorrelation picture gets genuinely more complicated at eight years than at three, though: rainfall's spatial clustering holds up in every single year, but SIF's is significant in only half, concentrated in years this study also calls drought or near-drought (2015, 2016, 2018, 2020). That points to something the three-year study couldn't have shown — SIF's spatial coherence across districts may itself strengthen under regional stress and fade during ordinary seasons, an additive finding from the expanded sample rather than one that simply confirms or contradicts the original.

Section 4.9's comparison against the one available official drought-declaration date (2018), after the boundary correction, plays out exactly the caution this Discussion keeps urging elsewhere: 2018 is this study's outlier year for the SIF-NDVI relationship on every measure computed, so NDVI crossing its decline threshold slightly before SIF in the corrected 2018 data fits that pattern rather than contradicting it. Section 1.1's original motivation treats a faster satellite-based indicator as a route to faster PMFBY payouts; this comparison shows the dominant gap in practice isn't SIF versus NDVI (a matter of days, in either direction depending on the year) but satellite-based monitoring of either kind versus the current declaration and payout process (a matter of roughly seven to eight weeks). None of that weakens the case for pursuing SIF further — a physiologically earlier indicator is worth chasing on its own terms, and RQ1's finding stands regardless of any single year's comparison — but it does mean the accurate version of this study's policy argument is "satellite monitoring generally, refined by SIF specifically, in most but not all years," not "SIF specifically, every year," as the primary lever on payout timing.

6. Limitations

- GOSIF, the SIF product used here, isn't a direct satellite retrieval — it's a statistically modeled reconstruction built from MODIS reflectance data, OCO-2 SIF soundings, and meteorological reanalysis, combined to give continuous spatiotemporal coverage. Since MODIS reflectance also underlies the NDVI product used for comparison, the two variables don't come from fully independent measurements at the input-data level. That's an inherent feature of the most widely available continuous SIF product, not a choice specific to this study, but it does mean the SIF-NDVI lag reported in Section 4.2 can't be fully attributed to independent physical signals — and this study makes no attempt to quantify or correct for that modeling dependency.
- The sample now runs eight years (2015–2023, excluding 2021), up from the original three, specifically to fix the small-sample limitation every earlier version of this paper carried. Even at eight years, only two (2015, 2018) meet this study's own rainfall-anomaly drought threshold (Section 3.5), so Section 4.2's drought-versus-normal comparison stays a two-versus-six split, not a balanced one — and one more drought year in the future could still shift the drought-group average considerably, given how few observations make it up.
- 2018 breaks this study's central SIF-leads-NDVI finding consistently, across every method used — threshold-crossing, cross-correlation, bootstrap, and the RQ3 declaration-timeline comparison — and this study lacks the crop-type, sowing-date, or within-season rainfall-timing data that would explain why 2018 in particular differs from the other seven years.
- The spatial correspondence between rainfall deficit and SIF stress (Section 4.5) got tested statistically via Pearson and Spearman correlation ($r = 0.567$, $\rho = 0.551$, both $p < 0.0001$), not left as a purely visual read; but with only eight independent study years and eight geographically adjacent districts, the 64 district-year observations aren't fully independent, so this correlation counts as corroborating evidence rather than a formally independent statistical confirmation. Section 4.8 puts a number on that directly via Moran's I, significantly positive ($p < 0.05$) for rainfall anomaly in all eight years but for mean SIF in only four — meaning the spatial clustering behind this correlation, and with it the effective independent sample size, varies by year instead of staying fixed.
- The low-SIF zones identified here haven't been checked against ground-level crop-stress or drought-impact reporting for the districts concerned.
- District-level rainfall anomaly was computed against one region-wide climatological baseline, not a per-district climatology.
- Section 4.9 compares SIF- and NDVI-based stress timing against the one well-documented official drought-declaration date I could find (Maharashtra, 31 October 2018). No comparably verifiable

declaration date turned up for any other study year, so this stays a single-year illustration, not a general finding across all eight — and as Section 4.9 details, even this single year's SIF-versus-NDVI ordering flipped after the Section 3.2 boundary correction, a reminder not to treat this comparison as more stable or precise than its sample size can actually support.

- The Google Earth Engine queries behind the NDVI, land-cover mask, and rainfall data (Sections 3.1–3.2) are now checked into version control as a runnable script (`src/acquisition/gee_data_acquisition.py`, using `earthengine-api`), rather than run only interactively — and this script produced all eight years of NDVI and rainfall data in this paper, including the five newly added years. Running it against the real Earth Engine service turned up two genuine bugs that a purely interactive workflow could easily have hit too and absorbed silently, without a code review ever catching them: a missing registered Cloud project (Earth Engine's current authentication requirement), and a MODIS-native-projection clip error from clipping a sinusoidal-grid image against a geographic-CRS boundary, fixed by dropping a redundant clip call. Both are logged in GA_Development_Log.md. GOSIF's clipping and every processing step after it remain fully scripted, executed, and reproducible as before.
- The exact numeric lag values in Section 4.2, and especially the ranking of which year shows the largest lag, depend on which lag-estimation method is used. The cross-correlation robustness check (Section 4.6) backs the weaker-but-real direction of the core finding — SIF's lag is never negative, in any of the eight years — but doesn't reproduce the threshold-crossing method's inter-annual ranking, and finds a clearly positive lag in only half the years instead of all of them. Section 4.7's bootstrap analysis puts a number on that: all 28 possible pairwise between-year comparisons of cross-correlation lag estimates have overlapping 95% confidence intervals, so the exact ranking isn't statistically distinguishable from noise even at this larger sample size. The robust result of this study is the qualitative one — SIF's decline is never later than NDVI's — while the exact day-count lag values, and any specific year-to-year ranking, should be read as method-dependent, wide-uncertainty estimates rather than precise, fixed quantities.

7. Conclusion

Across eight growing seasons in Marathwada, Maharashtra, Solar-Induced Fluorescence registers the onset of post-peak seasonal vegetation decline no later than NDVI, and measurably earlier in most years — a finding an independent cross-correlation method and a bootstrap resampling procedure both corroborate in direction, if not always in magnitude, and one that has now replicated on a sample more than twice the size of this study's original three-year version. There's no evidence, though, that drought conditions specifically widen this lag; if anything, the two years meeting this study's own drought threshold show a smaller average lag than the six normal years, the same direction the original, smaller sample found. Expanding the sample turned up two findings that qualify the original study rather than simply reinforce it. First, only two of the eight years studied actually meet this study's own rainfall-anomaly drought definition, meaning the original three-year sample was considerably more drought-heavy than Marathwada's typical year-to-year climate. Second, 2018 is a consistent, multiply-confirmed exception to this study's central finding — a near-zero or slightly reversed SIF-NDVI relationship shows up across every method used, including a direct reversal of the RQ3 declaration-timeline ordering originally reported for that year. Independent rainfall validation and spatial cross-referencing between SIF and precipitation data keep lending physical coherence to the district-level findings: a correspondence confirmed statistically (Pearson $r = 0.567$, Spearman $\rho = 0.551$, both $p < 0.0001$ — weaker than, but still highly significant relative to, the original smaller-sample estimate) and characterized further through a Moran's I spatial-autocorrelation diagnostic that itself shows SIF's spatial clustering is less consistent across years than rainfall's is. Checking against the one available official drought-declaration date

(2018) sharpens this study's policy framing toward satellite monitoring in general, with SIF as a specific but not universally reliable physiological refinement, rather than positioning SIF's edge over NDVI as a guaranteed, one-directional advantage. Several limitations — a still-unbalanced drought/normal group size, an unexplained year-specific exception, the spatial non-independence behind the district-level correlation, and the absence of ground validation — are stated directly, without resolving them beyond what the available data actually supports.

References

Didan, K. (2021). *MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061* [Data set]. NASA EOSDIS Land Processes Distributed Active Archive Center. https://doi.org/10.5067/MODIS/MOD13Q1.061

Down To Earth. (2025). *When crop insurance fails farmers: Why India's PMFBY scheme needs urgent reform*. https://www.downtoearth.org.in/agriculture/when-crop-insurance-fails-farmers-pmfby-needs-a-rethink

Economic and Political Weekly. (2018). Drought in Maharashtra [Editorial]. *Economic and Political Weekly*, 53(48). https://www.epw.in/journal/2018/48/editorials/drought-maharashtra.html

Zee News. (2018, October 31). *Maharashtra declares drought in 151 talukas in 26 districts*. https://zeenews.india.com/india/maharashtra-declares-drought-in-151-talukas-in-26-districts-2152281.html

Food and Agriculture Organization of the United Nations (FAO). (2015). *FAO GAUL (Global Administrative Unit Layers): Global Administrative Unit Layers 2015, Level 2* [Data set]. Google Earth Engine Data Catalog. https://developers.google.com/earth-engine/datasets/catalog/FAO_GAUL_2015_level2

Friedl, M., & Sulla-Menashe, D. (2022). *MODIS/Terra+Aqua Land Cover Type Yearly L3 Global 500m SIN Grid V061* [Data set]. NASA EOSDIS Land Processes Distributed Active Archive Center. https://doi.org/10.5067/MODIS/MCD12Q1.061

Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A., & Michaelsen, J. (2015). The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Scientific Data*, 2, Article 150066. https://doi.org/10.1038/sdata.2015.66

Li, X., & Xiao, J. (2019). A global, 0.05-degree product of solar-induced chlorophyll fluorescence derived from OCO-2, MODIS, and reanalysis data. *Remote Sensing*, 11(5), 517. https://doi.org/10.3390/rs11050517

Ministry of Agriculture & Farmers Welfare, Government of India. (2016). *Pradhan Mantri Fasal Bima Yojana (PMFBY)*. Retrieved from https://pmfby.gov.in